

Machine Learning Lab – Question 8

Question

Write a program to implement Linear Regression (LR) algorithm in Python.

Code

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([[1],[2],[3],[4],[5]])
y = np.array([2,4,5,4,5])

model = LinearRegression()
model.fit(X, y)
pred = model.predict(X)

print("Slope:", round(model.coef_[0], 4))
print("Intercept:", round(model.intercept_, 4))
print("R2 Score:", round(r2_score(y, pred), 4))
print("Prediction for x=6:", round(model.predict([[6]])[0], 4))
```

Output

```
Slope: 0.6
Intercept: 2.2
R2 Score: 0.6
Prediction for x=6: 5.8
```